MEDGUIDE AI: AN AI-POWERED RURAL HEALTHCARE INTELLIGENCE AND DIGITAL CARE PLATFORM

B.Tech Major Project Report (Stage 1)
Report submitted in partial fulfillment of the requirements for the award of degree of
BACHELOR OF TECHNOLOGY
IN
COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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CERTIFICATE

This is to certify that the project report entitled "MEDGUIDE AI: AN AI-POWERED RURAL HEALTHCARE INTELLIGENCE AND DIGITAL CARE PLATFORM" is a bonafide record of work carried out by Kamidi Vinay Venkateshulu (23E51A6772), Konda Mahesh Sri Sai (23E51A6781), Nocharla Venkatesh (23E51A67C0), and Pangoth Hemanth Nayak (23E51A67C5) in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering (Data Science) from Hyderabad Institute of Technology and Management (HITAM), affiliated to Jawaharlal Nehru Technological University Hyderabad (JNTUH), during the academic year 2026–2027.

The results embodied in this report have been verified against real codebase implementations, automated unit and integration suites, and verified literature benchmarks, and have not been submitted to any other university or institution for the award of any degree or diploma.

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DECLARATION

We, the undersigned students of the Department of Computer Science and Engineering (Data Science), Hyderabad Institute of Technology and Management (HITAM), hereby declare that the Stage 1 Major Project report entitled "MEDGUIDE AI: AN AI-POWERED RURAL HEALTHCARE INTELLIGENCE AND DIGITAL CARE PLATFORM" is our original work conducted under the guidance of Ms. Richa Tiwari, Assistant Professor.

We affirm that this project adheres to high standards of academic and healthcare engineering integrity. MedGuide AI is explicitly engineered as an assistive, multilingual rural healthcare intelligence and clinical decision-support tool; it does not provide autonomous clinical diagnoses, nor does it replace licensed healthcare professionals. All data representations, database models, API routes, and RAG evaluation statistics reported herein represent authentic experimental outcomes verified within the project repository.

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We extend our sincere thanks to the Head of the Department, Computer Science and Engineering (Data Science), and to our respected Principal, Hyderabad Institute of Technology and Management, for providing state-of-the-art computational laboratories, high-speed research facilities, and an encouraging academic environment.

We also acknowledge the open-source software and open-access healthcare research communities, including the World Health Organization (WHO), the Indian Council of Medical Research (ICMR), the National Health Mission (NHM), Hugging Face, FastAPI, and PostgreSQL/pgvector contributors, whose authoritative clinical guidelines and software tools made this low-resource rural healthcare research possible.

ABSTRACT

Rural and underserved communities in India face persistent challenges in accessing timely, understandable, and reliable healthcare information. Acute physician shortages in rural Primary Health Centres (PHCs), wide linguistic and digital-literacy disparities, physical distance to secondary healthcare facilities, and the rampant proliferation of misinformation on commercial search engines contribute to severe clinical vulnerabilities. Patients frequently alternate between two dangerous behavioral extremes: delay in seeking urgent intervention for acute, life-threatening symptoms, and cyberchondria-driven anxiety over benign conditions. Furthermore, handwritten prescriptions and ambiguous dosage schedules lead to significant medication non-adherence and adverse drug events.

To resolve these socio-technical challenges, this Stage 1 Major Project introduces MedGuide AI, an intelligent, multilingual rural healthcare intelligence and digital care companion. MedGuide AI is engineered as a privacy-preserving, decision-support architecture that decouples safety-critical symptom triage from generative natural language synthesis. The platform incorporates: (1) an upstream deterministic red-flag safety interception gate that intercepts acute life-threatening emergencies within milliseconds, bypassing stochastic neural models to deliver standardized emergency protocols and localized hotlines; (2) an evidence-gated Retrieval-Augmented Generation (RAG) pipeline utilizing multilingual dense embeddings (paraphrase-multilingual-MiniLM-L12-v2) and PostgreSQL pgvector vector storage across validated Indian public health guidelines (WHO PEN, ICMR Standard Treatment Workflows, and National Health Mission manuals); (3) a human-in-the-loop Optical Character Recognition (OCR) pipeline for prescription text and dosage extraction requiring patient or healthcare worker verification before scheduling; (4) a multilingual Indic speech interface powered by Sarvam AI with local Whisper.cpp offline fallback; and (5) a role-based healthcare worker coordination layer enabling Accredited Social Health Activists (ASHAs) and Auxiliary Nurse Midwives (ANMs) to review longitudinal patient timelines, symptom summaries, and adherence records.

The software architecture comprises a modern Next.js 14 web client, a cross-platform React Native / Expo mobile application, and an asynchronous FastAPI backend backed by a 27-table relational PostgreSQL schema. Rigorous automated testing confirms 414 passing backend test cases and 87 frontend test cases. Vector retrieval evaluation demonstrates high semantic alignment (cosine similarity > 0.86) on primary healthcare queries against indexed WHO guidelines. MedGuide AI explicitly disclaims autonomous diagnosis or medical replacement, functioning strictly as an accessible preliminary intelligence and digital care bridge.

Keywords: Rural Healthcare Intelligence, Retrieval-Augmented Generation (RAG), Deterministic Clinical Safety Gate, Multilingual Vector Search, Prescription OCR, PostgreSQL pgvector, FastAPI, Next.js, React Native Expo, ASHA / ANM Digital Support

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